

Test: 2022 Elementary Regional Science

Points: 36 points

Question 1 of 36

Exam Body

All matter has both physical and chemical properties. Which of the following physical properties refers to how much space is occupied by an object?

- ☐ A) density
- ☐ B) volume
- ☐ C) mass
- ☐ D) weight

Question 2 of 36

Exam Body

Intensive properties do not depend on the amount of matter involved. Which intensive property below is defined as the temperature at which a solid begins to turn into a liquid?

- ☐ A) freezing point
- ☐ B) boiling point
- ☐ C) vanishing point
- ☐ D) melting point

Question 3 of 36

Exam Body

One concept of motion is often explained in terms of Newton's three laws. Which of the following can cause an object to accelerate when it acts on that object?

- ☐ A) force
- ☐ B) mass
- ☐ C) density
- ☐ D) luminosity

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Exam Body

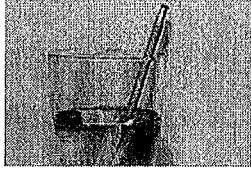
One of the terms below is related to but different from displacement. Which of the following scalar quantities in physics is defined as how much ground an object has covered during its motion?

- ☐ A) expanse
- ☐ B) metric
- ☐ C) distance
- ☐ D) length

Question 5 of 36

Exam Body

A classic example of one change is shown in the photograph, where the pencil seems to be broken. What change below is caused when light passes from one medium to another?



- ☐ A) reflection
- ☐ B) deflection
- ☐ C) infraction
- ☐ D) refraction

Question 6 of 36

Exam Body

Heating water in a kettle is an example of one kind of heat transfer. In which process below is heat transferred by the movement of heated air, water, or other fluids?

- ☐ A) conduction
- ☐ B) convection
- ☐ C) diffusion
- ☐ D) thermal radiation

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Exam Body

A unit used in electricity takes its name from an 18th- and 19th-century Italian physicist. Which of the following is the unit of electrical potential and is often used in connection with batteries?

- ☐ A) volt
- ☐ B) ampere
- ☐ C) ohm
- ☐ D) watt

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Exam Body

One aggregate material includes pebbles, cobbles, and boulders. Which of the following materials is used to make concrete and to cover roads and driveways?

- ☐ A) macadam
- ☐ B) asphalt
- ☐ C) bitumen
- ☐ D) gravel

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Exam Body

Modern chemical fertilizers used to replenish the soil usually contain three main elements. Which element below is NOT one of those three elements?

- ☐ A) potassium
- ☐ B) nitrogen
- ☐ C) oxygen
- ☐ D) phosphorus

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Exam Body

Match the animal with the type of invertebrate it is.

x. shrimp	1. arachnid
y. snail	2. crustacean
z. scorpion	3. gastropod

- ☐ A) x-3, y-1, z-2
- ☐ B) x-1, y-3, z-2
- ☐ C) x-2, y-1, z-3
- ☐ D) x-2, y-3, z-1

Question 11 of 36

Exam Body

Depending on their denomination, Christians celebrate Christmas in December or January. During which season do Christians in New Zealand celebrate Christmas?

- ☐ A) autumn
- ☐ B) summer
- ☐ C) winter
- ☐ D) spring

Question 12 of 36

Exam Body

In which stage of the water cycle does sunlight warm surface water, causing it to become a gas and rise into the atmosphere?

- ☐ A) evaporation
- ☐ B) condensation
- ☐ C) precipitation
- ☐ D) percolation

Question 13 of 36

Exam Body

The discoveries that one astronomer called celestial harmonies are now called laws. Which of the following came up with three major laws about the motion of planets?

- ☐ A) Nicolaus Copernicus
- ☐ B) Galileo Galilei
- ☐ C) Tycho Brahe
- ☐ D) Johannes Kepler

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Exam Body

The third largest planet by mass in the solar system was discovered in 1846. Which planet is named for the Roman god of the sea?

- ☐ A) Jupiter
- ☐ B) Saturn
- ☐ C) Neptune
- ☐ D) Uranus

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Exam Body

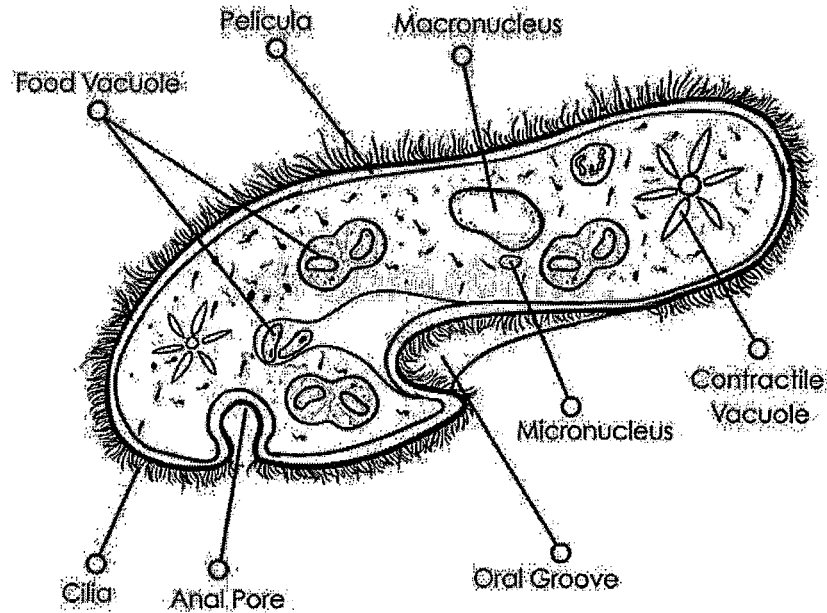
One solar phenomenon often emits electron streams and X-rays. Which of the following occurs when the corona suddenly gets much brighter?

- ☐ A) solar flare
- ☐ B) solar wind
- ☐ C) solar cycle
- ☐ D) solar eclipse

Question 16 of 36

Exam Body

The diagram shows a microscopic, one-celled protozoan often used in laboratory studies. Which protozoa below is it?



- ☐ A) euglena
- ☐ B) paramecium
- ☐ C) amoeba
- ☐ D) trypanosoma

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Exam Body

Certain free particles are often associated with the endoplasmic reticulum. What particles are the site of protein synthesis in the cell?

- ☐ A) chloroplasts
- ☐ B) Golgi bodies
- ☐ C) ribosomes
- ☐ D) centrioles

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Exam Body

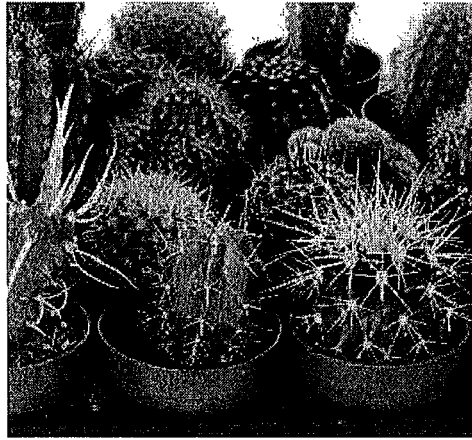
Ice land moss and manna are two types of a certain symbiotic organism. Which of the following organisms consists of an alga and a fungus?

- ☐ A) lichen
- ☐ B) rhizine
- ☐ C) cyanobacteria
- ☐ D) mistletoe

Question 19 of 36

Exam Body

The photo shows some of the nearly 2,000 species of a mostly Western Hemisphere plant family. Which of the following include cholla, prickly pear, and saguaro?



- ☐ A) acanthi
- ☐ B) legumes
- ☐ C) cacti
- ☐ D) saxifrages

Question 20 of 36

Exam Body

Match the part of a plant with its function.

x. sepal	1. protect developing reproductive structures
y. stem	2. support the buds, flowers, leaves, and fruits
z. style	3. support the stigma

- ☐ A) x-2, y-3, z-1
- ☐ B) x-1, y-2, z-3
- ☐ C) x-1, y-3, z-2
- ☐ D) x-3, y-1, z-2

Question 21 of 36

Exam Body

Insects go through metamorphosis. Which of the following is the third stage of complete metamorphosis?

- ☐ A) pupa
- ☐ B) embryo
- ☐ C) imago
- ☐ D) larva

Question 22 of 36

Exam Body

Humans go through various stages of development. Which of the stages below lasts from birth until someone normally begins to talk?

- ☐ A) adolescence
- ☐ B) puberty
- ☐ C) childhood
- ☐ D) infancy

Question 23 of 36

Exam Body

For humans, one function of a particular part of the body is to help you stay balanced when walking. Which of the following is a flexible column that stretches from the hips to the neck?

- ☐ A) coccyx
- ☐ B) femur
- ☐ C) spine
- ☐ D) sacroiliac

Question 24 of 36

Exam Body

You have — unless you've had them removed — small masses of tissue on your pharynx. Which of the following are often removed because they keep getting infected?

- ☐ A) adrenals
- ☐ B) tonsils
- ☐ C) lymphocytes
- ☐ D) leukocytes

Question 25 of 36

Exam Body

The American Dental Association recommends that you should do which of the following once a day to remove plaque from between your teeth?

- ☐ A) brush
- ☐ B) gargle
- ☐ C) rinse
- ☐ D) floss

Question 26 of 36

Exam Body

A unit used in technology is equal to 1 billion bytes. Which of the following units is it, often used as a selling point for smartphones?

- ☐ A) megabyte
- ☐ B) gigabyte
- ☐ C) petabyte
- ☐ D) kilobyte

Question 27 of 36

Exam Body

Zoologist Georges Cuvier was one scientist who paved the way for another kind of scientist. Which type of scientist studies fossils?

- ☐ A) paleontologist
- ☐ B) otologist
- ☐ C) teleologist
- ☐ D) gerontologist

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Exam Body

NASA chose six women in 1978 to train as astronauts. Which of the following became the first U.S. woman in space?

- ☐ A) Kathryn Sullivan
- ☐ B) Christa McAuliffe
- ☐ C) Sally Ride
- ☐ D) Christina Koch

Question 29 of 36

Exam Body

The scientist credited with saving France's wine industry also is noted for his work in medicine. Which scientist below developed a vaccine against rabies?

- ☐ A) Henri Becquerel
- ☐ B) André Malraux
- ☐ C) Pierre Curie
- ☐ D) Louis Pasteur

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Exam Body

Match the part of the body with the system of which it is a part.

x. aorta	1. digestive
y. stomach	2. endocrine
z. pancreas	3. circulatory

- ☐ A) x-2, y-3, z-1
- ☐ B) x-1, y-2, z-3
- ☐ C) x-3, y-1, z-2
- ☐ D) x-2, y-1, z-3

Question 31 of 36

Tiebreakers

Many solid objects have one property that comes into play when an object is deformed. Which property below allows deformed matter to return to its original shape?

- ☐ A) elasticity
- ☐ B) ductility
- ☐ C) solubility
- ☐ D) stability

Question 32 of 36

Tiebreakers

The least common type of coal is found mostly in northeastern Pennsylvania in the United States. Which type of coal below is also known as "hard coal"?

- ☐ A) lignite
- ☐ B) bituminous
- ☐ C) maceral
- ☐ D) anthracite

Question 33 of 36

Tiebreakers

Clouds are divided into high, medium, and low depending on the level of the atmosphere where they occur. Which of the following clouds is considered a high level cloud?

- ☐ A) stratus
- ☐ B) cirrus
- ☐ C) stratocumulus
- ☐ D) cumulus

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Tiebreakers

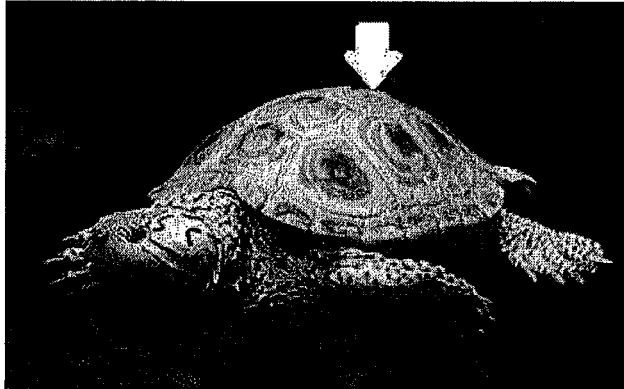
Amoebas often push out cytoplasm to create a protrusion that is used for movement or to trap prey. What name, meaning "false foot," is given to this protrusion?

- ☐ A) flagellum
- ☐ B) pseudopod
- ☐ C) cilium
- ☐ D) arthropod

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Tiebreakers

Turtles, tortoises, and terrapins have shells. To which part of the shell is the white arrow pointing in the photograph?



- ☐ A) carapace
- ☐ B) plastron
- ☐ C) girdle
- ☐ D) scutes

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Tiebreakers

A woman born in England who became a U.S. citizen faced opposition when she began attending Geneva Medical College in New York. Which of the following was the first woman to graduate as a doctor of medicine in the United States in modern times?

- ☐ A) Florence Nightingale
- ☐ B) Elizabeth Blackwell
- ☐ C) Rebecca Lee Crumpler
- ☐ D) Harriot Hunt